

Book 1 section 2.2: DSE multiple choice

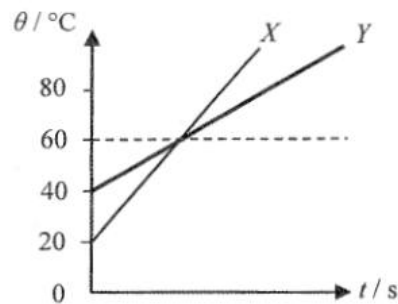
2025 Q1

1. An equal amount of energy is supplied to each of the substances P , Q , R and S of mass and specific heat capacity listed below. Which substance would have the largest rise in temperature?

	substance	mass / kg	specific heat capacity / $\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
A.	P	2	4200
B.	Q	3	2100
C.	R	4	900
D.	S	5	840

2015 Q2

2.



Two objects X and Y are made of the same material. They are heated separately by heaters of the same power. The graph shows the variation of temperature θ of X and Y with time t . What is the ratio of mass of X to that of Y ?

- A. 3 : 1
- B. 2 : 1
- C. 1 : 2
- D. 2 : 3

PP Q2

2. In the figure below, a training pool B is located next to the main pool A . The training pool B has a smaller area and is shallower. If the pools are under the sunlight at the same time, which of the following statements about the rise in the water temperature of the two pools is correct? Assume that the initial water temperatures of the pools are the same.



- A. The water temperature of training pool B rises faster because it is shallower.
- B. The water temperature of training pool B rises faster because it has a smaller surface area.
- C. The water temperature of main pool A rises faster because it is deeper.
- D. The water temperature of main pool A rises faster because it has a larger surface area.